→ Log-Likelihood of Dataset Given HMM

• Let T be the total # of timepoints in a sequence and $x_{l:r}$ be the observations from timepoint l to timepoint r. Then the likelihood of the sequence given the model is

$\mathcal{L} = p(x_{1:T}) = p(x_T, x_{1:T-1}) = p(x_T|x_{1:T-1})p(x_{1:T-1}) = p(x_T|x_{1:T-1})p(x_{T-1}|x_{1:T-2}) \dots p(x_1) = \prod_{t=1}^{T} p(x_t|x_{1:t-1})$, and the log-likelihood is

$\log \mathcal{L} = \log p(x_{1:T}) = \sum_{t=1}^{T} \log p(x_t|x_{1:t-1})$

• Let s_t denote the state at time t and $\underline{A}$ be the $k \times k$ state transition matrix learned by the model. Then

$$
\begin{aligned}
p(x_t|x_{1:t-1}) &= \sum_j p(x_t, s_t=j|x_{1:t-1}) \\
&= \sum_j p(x_t|s_t=j)p(s_t=j|x_{1:t-1}) \\
&= \sum_j p(x_t|s_t=j)\Big(\sum_i p(s_t=j|s_{t-1}=i)p(s_{t-1}=i|x_{1:t-1})\Big) \\
&= \sum_j p(x_t|s_t=j)\Big(\sum_i \underline{A}_{i,j}\, p(s_{t-1}=i|x_{1:t-1})\Big)
\end{aligned}
$$

$$
\begin{aligned}
p(s_t=j|x_{1:t}) &= \frac{p(s_t=j, x_{1:t})}{p(x_{1:t})} \\
&= \frac{p(x_t, s_t=j, x_{1:t-1})}{p(x_t, x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j, x_{1:t-1})}{p(x_t|x_{1:t-1})\,p(x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j|x_{1:t-1})p(x_{1:t-1})}{p(x_t|x_{1:t-1})\,p(x_{1:t-1})} \\
&= \frac{p(x_t|s_t=j, x_{1:t-1})p(s_t=j|x_{1:t-1})}{p(x_t|x_{1:t-1})} \\
&= \frac{p(x_t, s_t=j|x_{1:t-1})}{p(x_t|x_{1:t-1})} \qquad *
\end{aligned}
$$

$$
\begin{aligned}
\underbrace{\log p(x_t|x_{1:t-1})}_{\text{※log\_c}} &= \log \sum_j p(x_t, s_t=j|x_{1:t-1}) \\
&= \log \sum_j \exp(\underbrace{\log p(x_t, s_t=j|x_{1:t-1})}_{\text{※log\_fit (unnormalized)}}) \\
&= \log \sum_j \exp(\log p(x_t|s_t=j)p(s_t=j|x_{1:t-1})) \\
&= \log \sum_j \exp(\underbrace{\log p(x_t|s_t=j)}_{\text{※log\_B}} + \underbrace{\log p(s_t=j|x_{1:t-1})}_{\text{※log\_pred}})
\end{aligned}
$$

$\log p(x_t|s_t=j)$ is the emission log-likelihood for one timepoint t. osld provides a function to compute this

$$
\begin{aligned}
\log p(s_t=j|x_{1:t-1}) &= \log \sum_i \underline{A}_{i,j}\, p(s_{t-1}=i|x_{1:t-1}) \\
&= \log \sum_i \exp(\log \underline{A}_{i,j} + \log p(s_{t-1}=i|x_{1:t-1}))
\end{aligned}
$$

$\underline{A}_{i,j}$ is learned by the model

$\log p(s_{t-1}=i|x_{1:t-1})$ comes from the previous iteration, so for each time t, we must carry forward $\underbrace{\log p(s_t=i|x_{1:t})}_{\text{※log\_fit (normalized)}}$

$$
\begin{aligned}
\log p(s_t=i|x_{1:t}) &= \log \frac{p(x_t, s_t=j|x_{1:t-1})}{p(x_t|x_{1:t-1})} \quad \text{by } * \\
&= \log p(x_t, s_t=j|x_{1:t-1}) - \log p(x_t|x_{1:t-1}) \\
&= \text{log\_fit (unnormalized)} - \text{log\_c}
\end{aligned}
$$

• Note: For $t=1$, log_c $= \log p(x_1)$ because there's nothing to condition on yet. Similarly, log_pred $= \log(s_1=j) = \pi_j$, the j^{th} element of the initial state distr. $\vec{\pi} \in \mathbb{R}^k$

• Thus, the full log-likelihood of a dataset D given a model, $\log \mathcal{L}(D)$, can be solved for iteratively. Suppose there are b batches in the dataset, each containing q sequences. Then $\mathcal{L}(D) = \prod_{u=1}^{b}\prod_{v=1}^{q} \mathcal{L}(x_{u,v})$, where $x_{u,v}$ is the v^{th} sequence of batch u, $1 \le u \le b$, $1 \le v \le q$. So

$\underbrace{\log \mathcal{L}(D)}_{\text{※ full\_loglik}} = \log \prod_{u=1}^{b}\prod_{v=1}^{q} \mathcal{L}(x_{u,v}) = \sum_{u=1}^{b} \log \prod_{v=1}^{q} \mathcal{L}(x_{u,v}) = \sum_{u=1}^{b}\sum_{v=1}^{q} \log \mathcal{L}(x_{u,v})$, i.e. the sum of the log-likelihoods of all $b \cdot q$ sequences